Anuj Pai Raikar

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EDUCATION

MS Robotics Engineering, ***Worcester Polytechnic Institute,*** *MA, USA* **Aug 2022 - May 2024**

SKILLS

Programming & Libraries: C++, Python, CUDA, PyTorch, TensorFlow, NumPy, Scikit-learn, Pandas, MATLAB, OpenCV, NetworkX
Tools: Foxglove, Google Test, RViz, Git, tf_transformations, ONNX, Gazebo, Unreal Engine, valgrind, gdb, AWS, Docker, Azure

WORK EXPERIENCE

AIML Software Engineer, Vizzhy Inc.*, TX* *August 2025 - Present*
• Engineering **LLM**-powered **app** that uncover patient health insights by integrating multi modal data streams, emphasizing scalability, accuracy, and deployment in real-world clinical settings.
• Designed and deployed a **GPU**-accelerated, low-latency **ML** based speech-to-text pipeline with VAD and **Open AI** Whisper, enabling accurate, real-time clinical transcription on **embedded edge devices** integrating audio, text, and images for reliable patient health logging.
• Tested and profiled ML pipelines, resolving latency and accuracy issues to ensure reliable, real-time performance in production environments.

ML Engineer Intern, Cybic Tech Inc, TX *April 2025 - July 2025*
• Working on ***machine learning*** based order time-series forecasting for health diagnosis tests using **Python, Pandas, AWS S3, SageMaker.**
• ***Data analysis, ETL*** & subsequent ***development and tuning*** of predictive ML models on client defined success criteria for suite of products.

Autonomy Engineer Intern, JLG Industries, MD *May 2024- Aug 2024*
• **Integrated *camera + LiDAR sensor fusion*** based perception stack; implementing ***AprilTag***-based docking procedure for **mobile robots**.
• Calibrated multi-frame robot transforms using **ROS2 TF** libraries to enable accurate goal pose computation for JLG AMR.
• Enhanced **Nav2** autonomy stack **(Hybrid A* global)** by modifying **behavior trees** for robust path planning under dynamic conditions.
• **Validated** docking pipeline in 50+ trials, achieving <3 cm positional error and reducing pose estimation errors across varied approach angles.

Machine Learning Researcher, Medical FUSION Lab, WPI, MA *Feb 2024 - May 2024*
• Explored CNNs to synthesize **ultrasound images** from **3D CT** Scans. Attained ***98% faster inference time*** as opposed to MUST simulations.
• Generated dataset by processing 3D CT scans in **NII format** using **MATLAB**. Experimented with **UNet**, Channel Attention & pix2pix **GAN**.

Robotics Perception Intern, DEKA Research and Development, NH *May 2023 - Aug 2023*
• Trained self-supervised learning model based on ***Vision Transformer*** (***LVM***) for ***semantic segmentation*** & anomaly detection in ***PyTorch***.
• Created data loader and processed large-scale internal image datasets in ***Python*** for multi-GPU training using **CUDA** and **Docker.**
• **TensorBoard** visualizations used to monitor model metrics and tune hyperparameters. ***mIoU = 34*** obtained using original parameters.

Robotics Graduate Researcher, MER Lab, WPI, MA *Sep 2023 – May 2024*
• Developed a **ROS** perception pipeline for autonomous object picking on a 7-DoF robotic arm (Franka Emika) with a variable-friction gripper.
• Processed **3D point clouds** with **PCL** for grasp points & benchmarked against **Fast RCNN** for unknown object grasp pose prediction & pick.
• Implemented **OMPL** RRTConnect and PRM with **Relaxed IK** solver for optimal and collision- free manipulator end-effector trajectories.

Associate Consultant, Oracle Financial Services Software, India *Sep 2020 - Mar 2022*
• Customized & tested banking software modules: Modified JavaScript and ***PL/SQL*** base packages according to client functional requirement.
• **SonarQube** for debugging & code quality assessment. ***Issue Tracking, Version Control*** with **JIRA, Git**. ***SOAP XML*** for Web Services tests.

PROJECTS

LiDAR Obstacle Detection — *C++, KD-Tree, PCL, Eigen, Boost*
• Built a real-time LiDAR-based obstacle detection pipeline in **C++17** by implementing voxel grid filtering, **RANSAC** plane segmentation, and Euclidean clustering using KD-Trees; optimized for performance and accuracy in highway driving scenarios.
• Processed real-world and simulated point cloud data to generate structured 3D obstacle detections with pose aligned bounding boxes, contributing to safer autonomous navigation and efficient scene understanding.

Sensor fusion for Pose Estimation of Quadrotor — *Python, MatPlotLib*
• Applied Complementary, Madgwick, and Unscented Kalman Filters to 6DoF IMU data from ArduIMU to estimate accurate aerial vehicle pose and benchmarked the results against ground truth data from a VICON Motion Capture system.

Perception & Navigation Stack for Autonomous drone - *Python, OpenCV, PyTorch, Path Planning, ONNX, TensorRT*
• Integrated geometric & learning-based **perception** techniques for object detection aboard **DJI Tello** paired with **NVIDIA Jetson Nano**.
• 100% collision free ***motion planning*** through tracks using 6-DoF ***pose estimation*** and **3D RRT* path planner** with quintic trajectory.
• Applied ***sim2real, multi-threading*** and ***TensorRT*** techniques for **2.5x faster inference** from neural networks for faster track navigation.
• Trained ***U-Net*** on synthetic ***domain randomized*** dataset for segmentation and ***SPyNet*** for ***deep optical flow*** based amorphous gap detection.

Real-Time Traffic Light Control and Vehicle Simulation - *C++, Multithreading, CMake, Make*
• Implemented a traffic **simulation** with vehicles and traffic lights, where cars wait at intersections until the light turns green before proceeding.
• Used **C++11 multithreading**, **mutexes,** and condition variables to synchronize vehicle movements and traffic light changes concurrently.
• Applied **object-oriented design** and create a graphical interface to visualize the traffic flow and light states in real time.